Chapter 7: System control engineering basic labs

Reporting Assignment 1.1

- The angular natural frequency (ω_0) is given by,

$$\omega_0 = \sqrt{\frac{k}{m}}$$

For $m = 1.1kg$ and $k - 1.2 \times 10^3 N/m$, and there is no damper,

$$\omega_0 = \sqrt{\frac{1.2 \times 10^3}{1.1}} s^{-1} = 33.02Hz \approx 33Hz$$

The natural frequency ($\omega = \omega_0/2\pi$) is given by,

$$\omega = \frac{33.02}{2\pi} = 5.26Hz \approx 5.3Hz$$

- No oscillations are produced when $\gamma \geq 1$. Since,

$$\gamma = \frac{c}{2m\omega_0}$$

$\gamma \geq 1$ when $c \geq 2m\omega_0$. Therefore, for this problem,

$$c \geq (2)(1.1)(5.26)$$

$$c \geq 11.56 \approx 11.6Ns/m$$

- For $c > 0$, $k > 0$, the solution (x) is given by,

$$e^{\frac{-c}{2m}t} Ccos(\frac{\sqrt{4mk - c^2}}{2m}t + \phi)$$

Considering the first part of the two multipliers $e^{\frac{-c}{2m}t}C$, taking the limit as time goes to infinity,

$$\lim_{t \to \infty} Ce^{\frac{-c}{2m}t} = Ce^{\frac{-c}{2m}\infty} = 0$$

Hence we can say that as enough time passes, displacement reaches zero for a regular spring-mass damper system.

Reporting Assignment 1.2

For $m = 1$, the matrix A can be found to be,

$$A = \begin{bmatrix} 0 & 1 \\ -k & -c \end{bmatrix}$$

The characteristic equation given by $\det(A - \lambda I)$ can be found to be,

$$\lambda^2 + c\lambda + k = 0$$

- For characteristic roots $\lambda_1 = 2i$ and $\lambda_2 = -2i$, the two characteristic questions below can be formed,

  $$-4 + c(2i) + k = 0$$

  $$-4 - c(2i) + k = 0$$

  These equations can now be simultaneously solved to get $c = 0$ and $k = 4$. From a physics perspective, this suggests that for these values of λ, there is no damping force but there is a spring force in the direction opposite to displacement.

- For characteristic roots $\lambda_1 = -1+2i$ and $\lambda_2 = -1-2i$, the two characteristic questions below can be formed,

  $$(-3 - 4i) + (-1 + 2i)c + k = 0$$

  $$(-3 + 4i) - (1 + 2i)c + k = 0$$

  These equations can now be simultaneously solved to get $c = 2$ and $k = 5$. From a physics perspective, this suggests that for these values of λ, there is both a damping force and a spring force in the direction opposite to displacement.

- For characteristic roots $\lambda_1 = -1$ and $\lambda_2 = 1$, the two characteristic questions below can be formed,

  $$1 - c + k = 0$$

  $$1 + c + k = 0$$

  These equations can now be simultaneously solved to get $c = 1$ and $k = -2$. From a physics perspective, this suggests that for these values of λ, there is a damping force opposing motion and a spring force that is in the same direction as motion (unusual). This would generally showcase an unstable system as energy is being "added" continuously.

- For characteristic roots $\lambda_1 = 1 + 2i$ and $\lambda_2 = 1 - 2i$, the two characteristic questions below can be formed,

  $$(-3 + 4i) + (1 + 2i)c + k = 0$$

  $$(-3 - 4i) + (1 - 2i)c + k = 0$$

  These equations can now be simultaneously solved to get $c = -2$ and $k = 5$. From a

physics perspective, this suggests that for these values of λ, there is a damping force in the direction of motion (effectively acting as a speeding up force) and there is a spring force in the direction opposite to displacement. This would also show an unstable system.

- For characteristic roots $\lambda_1 = -1$ and $\lambda_2 = -2$, the two characteristic questions below can be formed,

$$1 - c + k = 0$$

$$4 - 2c + k = 0$$

These equations can now be simultaneously solved to get $c = 3$ and $k = 2$. From a physics perspective, this suggests that for these values of λ, there is both a damping force and a spring force in the direction opposite to displacement.

Reporting Assignment 1.3

Using matrices, we can write the secular equation from the basic equation of motion using A in the following steps,

$$\frac{dx}{dt} = A\mathbf{x}$$

Since, A $= \begin{bmatrix} 0 & 1 \\ -k/m & -c/m \end{bmatrix}$, and $\mathbf{x} = \begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix}$ we have the equation,

$$\frac{d\mathbf{x}}{dt} = \begin{bmatrix} 0 & 1 \\ \frac{-k}{m} & \frac{-c}{m} \end{bmatrix} \begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix}$$

Solving this, we get the differential equation,

$$\frac{d^2x}{dt} + \frac{c}{m}\frac{dx}{dt} + \frac{k}{m}x = 0$$

Multiplying both sides by m, we get

$$m\frac{d^2x}{dt} + c\frac{dx}{dt} + kx = 0$$

This is the same as equation (1.3), which can be used to find the characteristic equation given below by substituting $x = e^{\lambda t}$,

$$m\lambda^2 + c\lambda + k = 0$$

The above equation exactly matches equation (1.4).

Reporting Assignment 1.4

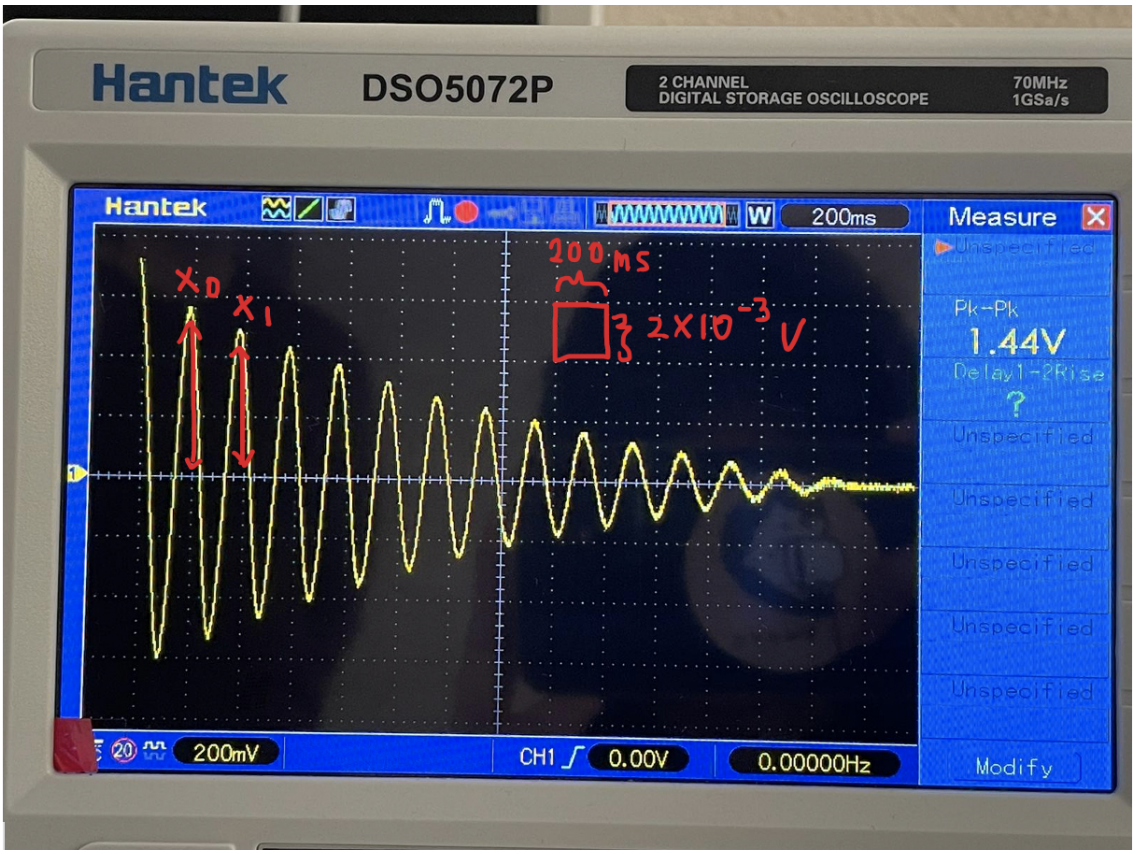


Figure 1: Oscilloscope time-history waveform

1. From Figure 1, the period (T) of the graph can be found to be 200 ms. Since $T = \frac{2\pi}{\omega_0}$, we get the natural angular frequency (ω_0) as,

$$\omega_0 = \frac{2\pi}{200 \times 10^{-3}} = 31.4Hz$$

From the formula for natural angular frequency ($\omega_0 = \sqrt{\frac{k}{m}}$) and mass ($m = 1.1kg$), we can get the value of k as,

$$k = \omega_0^2 m = (31.4)^2(1.1) = 1085.656 \approx 1085.7N/m$$

2. It was found that the damping ratio (γ) is related to logarithmic decrement (δ) in the following way (Inman & Singh, 2014),

$$\gamma = \frac{\delta}{\sqrt{\delta^2 + (2\pi)^2}}$$

where $\delta = \ln \frac{x_0}{x_1}$ and x_0, x_1 are amplitudes of two consecutive positive or negative peaks.

Calculating the above values from Fig 1 we get, $x_0 = (2.8)(10)(2 \times 10^{-3}) = 5.6 \times$

$10^{-2}mm$ and $x_1 = (2.5)(10)(2 \times 10^{-3}) = 5.0 \times 10^{-2}mm$. Therefore, $\delta = \ln \frac{x_0}{x_1} = \ln \frac{5.0}{5.6} = 0.113$. From this, we can get,

$$\gamma = \frac{0.113}{\sqrt{0.113^2 + (2\pi)^2}} = 0.018$$

We already know that $\gamma = \frac{c}{2m\omega_0}$. Rearranging the equation, we get the damping coefficient (c) as

$$c = 2m\omega_0\gamma = 2(1.1)(31.4)(0.018) = 1.246 \approx 1.25Ns/m$$

3. The eigenvalues λ_1 and $\lambda 2$ can be calculated with the following equation,

$$\lambda_{1,2} = \frac{-c \pm \sqrt{c^2 - 4mk}}{2m}$$

$$\lambda_{1,2} = \frac{-1.25 \pm \sqrt{(1.25)^2 - 4(1.1)(1.086 \times 10^3)}}{2(1.1)} = -0.567 \pm 31.4i$$

The two eigenvalues are: $\lambda_1 = -0.567 + 31.4i$ and $\lambda_2 = -0.567 - 31.4i$.

Reporting Assignment 1.5

1. The state equation for this system is given by the general equation,

$$\frac{d}{dt}\begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ \frac{-k}{m} - K_p & \frac{-c}{m} - K_d \end{bmatrix} \begin{bmatrix} x \\ \frac{dx}{dt} \end{bmatrix}$$

Solving this, we get the differential equation,

$$\frac{d^2x}{dt} + (\frac{c}{m} + K_d)\frac{dx}{dt} + (\frac{k}{m} + K_p)x = 0$$

For this system, the state equation is given as,

$$\frac{d^2x}{dt} + (1.14 + K_d)\frac{dx}{dt} + (9.87 \times 10^2 + K_p)x = 0$$

2. The Block Diagram would look like Figure 2

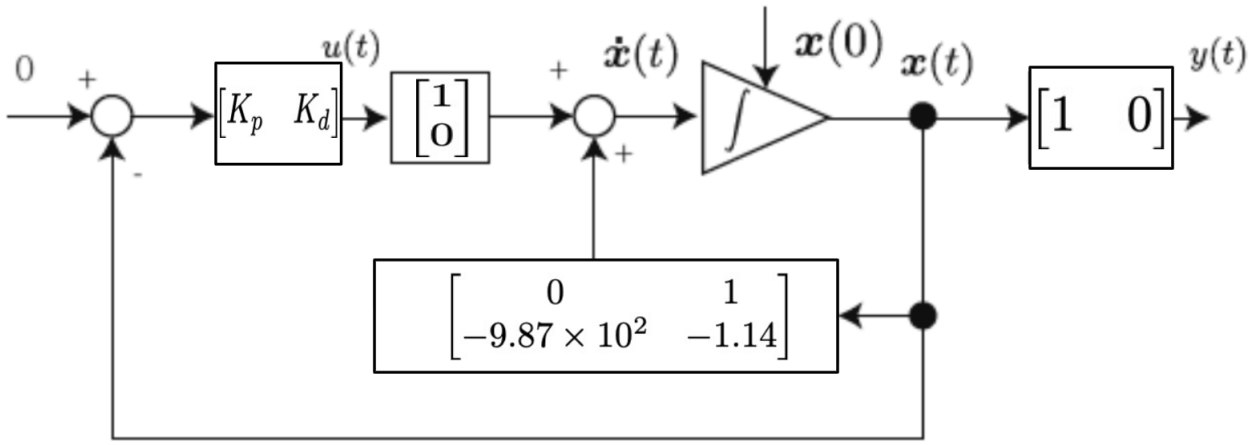


Figure 2: Block Diagram

This diagram is similar to Fig 1.2 in the textbook on page 40, where where $A = \begin{bmatrix} 0 & 1 \\ -9.87 \times 10^2 & -1.14 \end{bmatrix}$, $K = \begin{bmatrix} K_p & K_d \end{bmatrix}$, $B = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 \end{bmatrix}$.

Reporting Assignment 2.1

Figure 4 shows a diagram of displacement versus time with the phase difference (ϕ) and the amplitude ($\frac{a_i}{\sqrt{(k-m\omega^2)^2+c^2\omega^2}}$) clearly marked. Plotting the gain calculated from X_p using a Laplace transform with frequency will create a bode diagram that looks like Figure 5. The peak of the frequency response would indicate the resonance frequency of the spring-mass damper system.

Reporting Assignment 2.2

Graph 1 shows the Frequency (Hz) versus Gain (Output Amplitude/Input Amplitude) and the Frequency (Hz) versus Phase Difference (ms) graph measured on the oscilloscope in the laboratory. The excitation amplitude (input amplitude) was kept constant and frequency was varied from 4 Hz to 6 Hz with intervals of 0.2 Hz to plot a bode diagram.

Reporting Assignment 2.3

Graph 2 shows the Frequency (Hz) versus Gain (Output Amplitude/Input Amplitude) and the Frequency (Hz) versus Phase Difference (ms) graph measured on an AC voltage source in the laboratory. The excitation amplitude (input amplitude) was kept constant and frequency was varied from 8 Hz to 18 Hz with intervals of 1 Hz to plot a bode diagram.

Reporting Assignment 2.4

The calculated values of the program are shown in Figure 3. Comparing these with Graphs 1 and 2 , the following observations can be made:

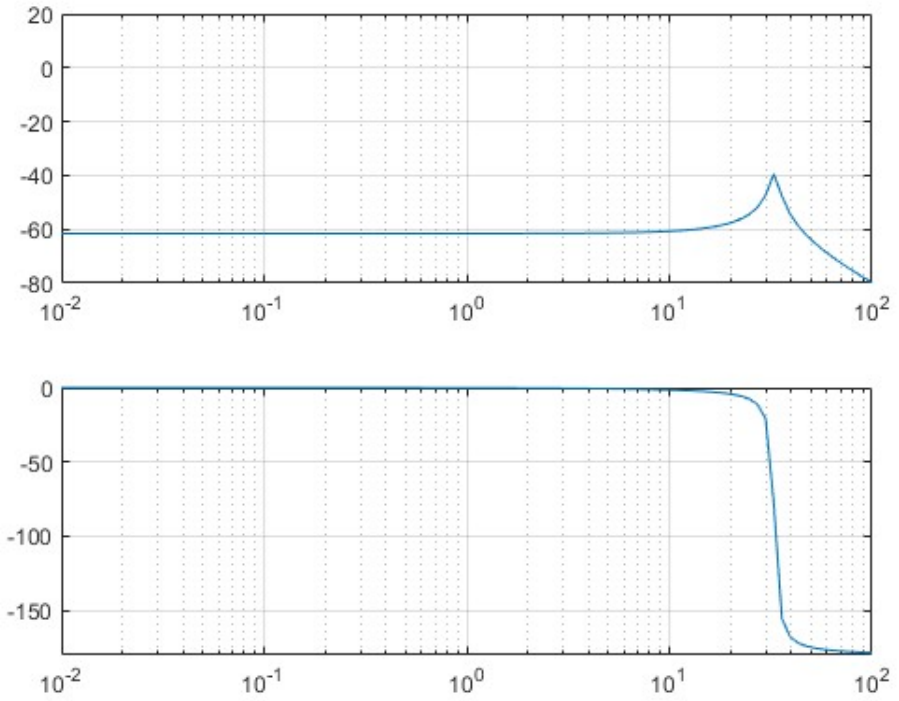


Figure 3: Program Plot (no axis labels in code)

Both Figure 3 and Graph 1 have a similar general shape with a slow initial growth, spiking up to a maximum (peak) and then a decay. As compared to the steepness of the spike in Figure 3 and Graph 1, Graph 2 has a relatively rounded maximum and not a clear, distinguishable peak. Factors that may affect the steepness of the peak include limitations in Capacitance and Inductance (usually frequency-dependent), Bandwidth (the frequency range where an AC source can provide a relatively constant voltage), and other component limitations. An AC source has a limited frequency range where it can maintain constant voltage, as opposed to a DC source which can keep voltage constant regardless of frequency which can exacerbate this difference.

Since the values plotted in the program are different to the variations of frequency and amplitude measured in the laboratory, the value of Gain at the peak is different. Since the program has no units or labels on the x and y axes, it can be assumed that the y-axis is in decibels (db). Whereas, the Gain values in Graphs 1, 2 are dimensionless.

Reporting Assignment 2.5

(a) The poles (K_p, K_i, K_d) are varied with using the following values in Practice Assignment 2.9: (500, 200, 500), (500, 0, 500), (500, 200, 0), (0, 200, 0), (0, 200, 500). Using this assignment, the following could be determined about the relationship between poles and responses: When K_p is set to zero, there is an initial dip in the graph, when K_i is set to zero, the graph never reaches its target value, and when K_d is set to zero, we can see sinusoidal behaviour at the beginning of the graph before it asymptotically reaches the target 1. When comparing the plots of the eigenvalues on the complex plane to the graphs, we can see that when K_i is set to zero, there are no imaginary eigenvalues.

This can help us establish the relationship between pole placement and frequency response by suggesting that:

i. K_p: The proportionality constant causes the frequency to reach the target with a constant gain factor. By increasing or decreasing this constant, you can increase the rate at which the target it reached.

ii. K_i: The integration constant is a function of the area under the target and the frequency response, which is sought to be minimized to reach the target. The integration constant is essential at making the frequency response reach its target. In the trial where both K_p and K_d were zero, the frequency response reached the target, but only in the trial with $K_i = 0$, the target was not reached.

iii. K_d: The differential constant is used to reduce the oscillations made by the frequency response when achieving the target. When K_d is set to zero, the integration constant has to detect a negative area before it can reduce the frequency and match the target. The differential constant measures the rate of change of frequency and slows down the speed of increasing frequency when the target is close. This gives a smoother constant curve as opposed to an oscillating one.

Conclusion

In conclusion, this lab allowed us to explore the field of Control Systems by altering m (mass), c (damping coefficient), k (spring constant) of a spring-mass-damper system, frequency response and PID control. In the first week, we observed the behaviour of a spring-mass damper system using simulations. By determining a mathematical equation, we calculated the c, k values of the real-life laboratory equipment using an oscilloscope that measured displacement using sensors (in terms of change in Voltage). In the second week we used both simulations and an oscilloscope with physical laboratory equipment to determine the impact of K_p (proportionality constant), K_d (differentiation constant) and K_i (integration constant) on the frequency response of spring-mass damper system run with a DC voltage source and an AC voltage source.

References

Inman, D. J., & Singh, R. C. (2014). *Engineering vibration*. Pearson Education Limited.